

# Plan

- Directions of Fea(LPP)
- Extreme directions of Fea(LPP)
- Representation Theorem for Fea(LPP)
- Necessary and sufficient conditions for the existence of optimal solutions
- Optimal solutions in atleast one corner point

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since  $\mathbf{x} + \alpha(\gamma \mathbf{d}) = \mathbf{x} + (\alpha\gamma) \mathbf{d} \in S$  for all  $\alpha > 0$



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are all equal as directions of  $\text{Fea}(LPP)$
- Whereas  $\mathbf{d}_1 = [1, 1]^T, \mathbf{d}_2 = [1, 0]^T$  give two **distinct directions**

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- The set of all **directions** of  $S = \text{Fea}(LPP)$  is a **convex set**

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- These are the only **extreme directions** of  $S$

- **Theorem:**

If  $S = \{\mathbf{x} \in \mathbb{R}^n : A_{m \times n} \mathbf{x} \leq \mathbf{b}, \mathbf{x} \geq \mathbf{0}\}$  is nonempty, then  $S$  has at least one **extreme point**



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- The theorem works for  $Fea(LPP)$  because of the **non negativity constraints**, and  $Fea(LPP)$  has  **$n$  LI**, **defining hyperplanes**

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- **Definition:** Given  $S$ , a nonempty subset of  $\mathbb{R}^n$ , and  $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k \in S$   
 $\sum_{i=1}^k \lambda_i \mathbf{x}_i$ , is called a convex combination of  $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k$  where  $0 \leq \lambda_i \leq 1$  for all  $i = 1, 2, \dots, k$ , and  $\sum_{i=1}^k \lambda_i = 1$

- All possible **convex combinations** of **two distinct points** gives a **straight line segment** with those two points as boundary points.



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- **Result:** Given  $\emptyset \neq S \subset \mathbb{R}^n$ ,  $S$  is a **convex set**  $\Leftrightarrow$  for all  $k \in \mathbb{N}$ ,  
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- **Theorem: (Representation Theorem)**

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$$[1, \frac{1}{2}]^T = \frac{5}{12}[2, 0]^T + \frac{3}{24}[0, 2]^T + \frac{1}{6}[1, \frac{3}{2}]^T + \frac{7}{24}[0, 0]^T$$

- Since  $D'$ , the set of **distinct** directions of  $S$  (if it is nonempty) is **bounded** (  $\mathbf{d} \geq \mathbf{0}$  and  $\sum_{i=1}^n d_i = 1$  )  
so any  $\mathbf{d} \in D'$  is a **convex combination** of the **extreme points** of  $D'$

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